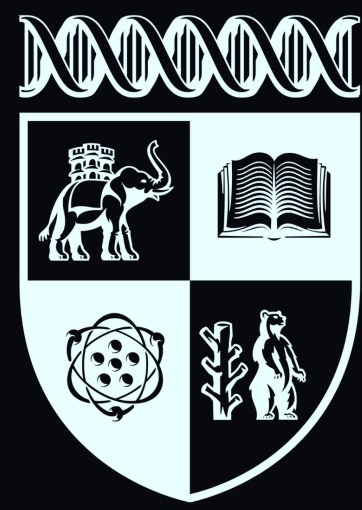




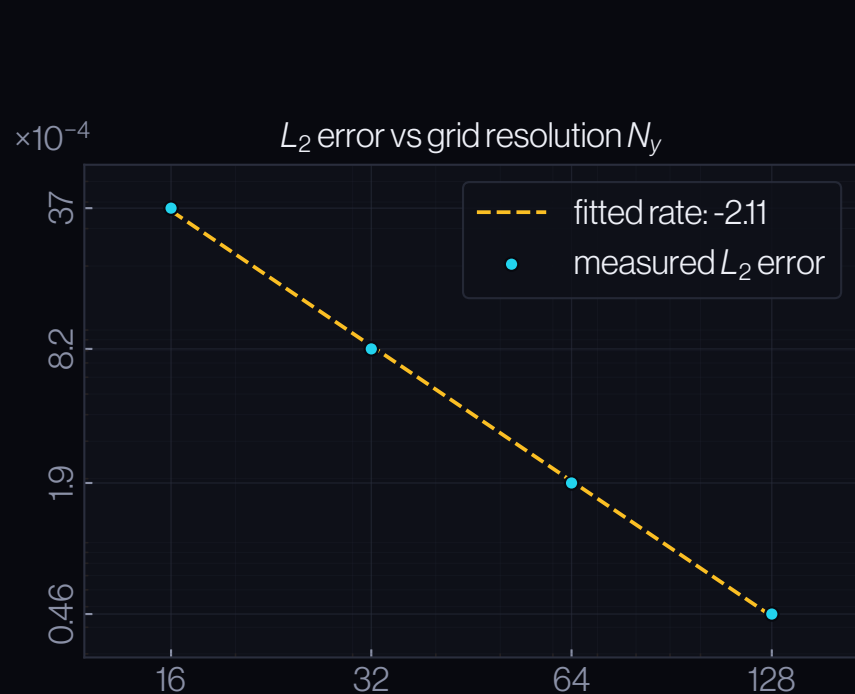
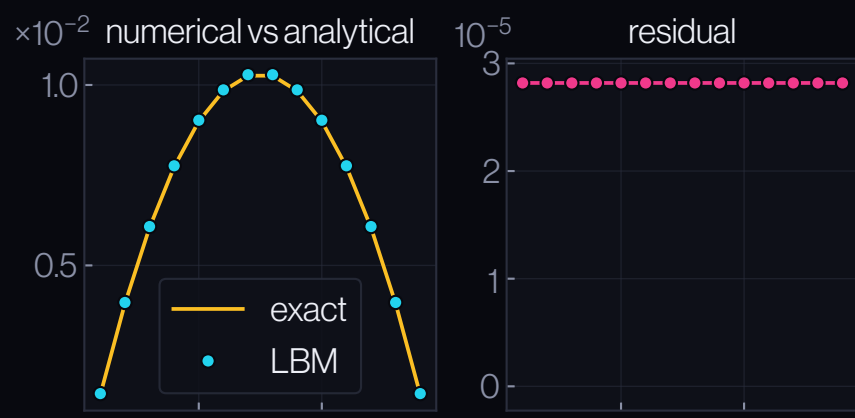
Adjoint lattice Boltzmann topology optimization

.01

motivation

Engineering is slow because it is iterative. Draft, simulate, refine, repeat. What if the rules could be encoded, and the structure generated instead of designed by hand?

The aim was to build a solver that designs 2D pipes given only the inlets, the outlets, and constraints on how much material can be used. The objective is to minimize energy dissipation.



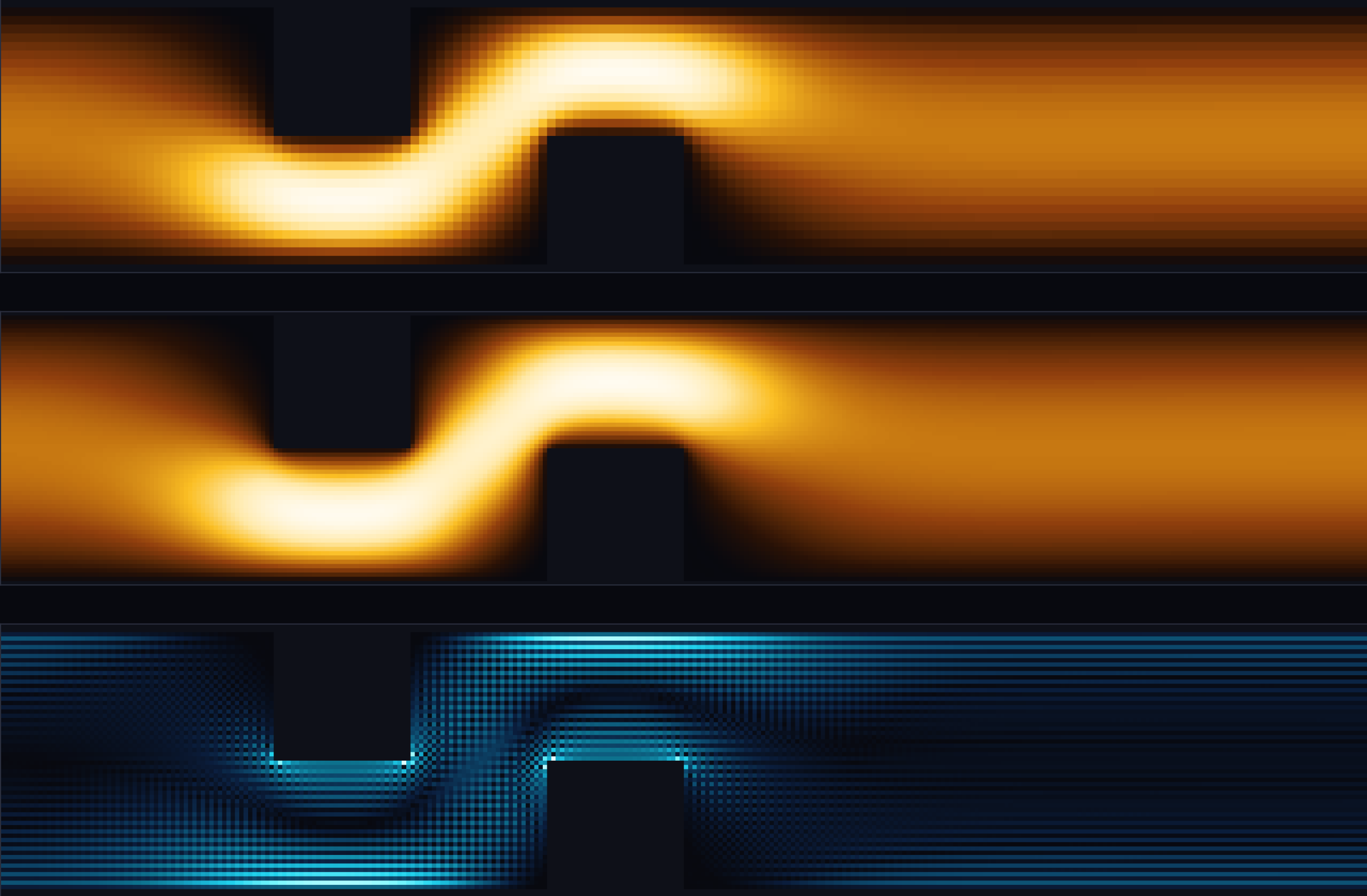
.02

analytical verification

The lattice Boltzmann method replaces Navier–Stokes mesh solvers by tracking particle distribution functions streaming and colliding on a discrete lattice grid.

We compared our simulated velocities to the exact solution for Poiseuille flow between two plates. The error falls fourfold every time the grid doubles – second-order convergence, the expected rate, confirming the implementation is correct.

methods
D2Q9 LBM, BGK collision, Zou–He pressure boundaries, half-way bounce-back walls



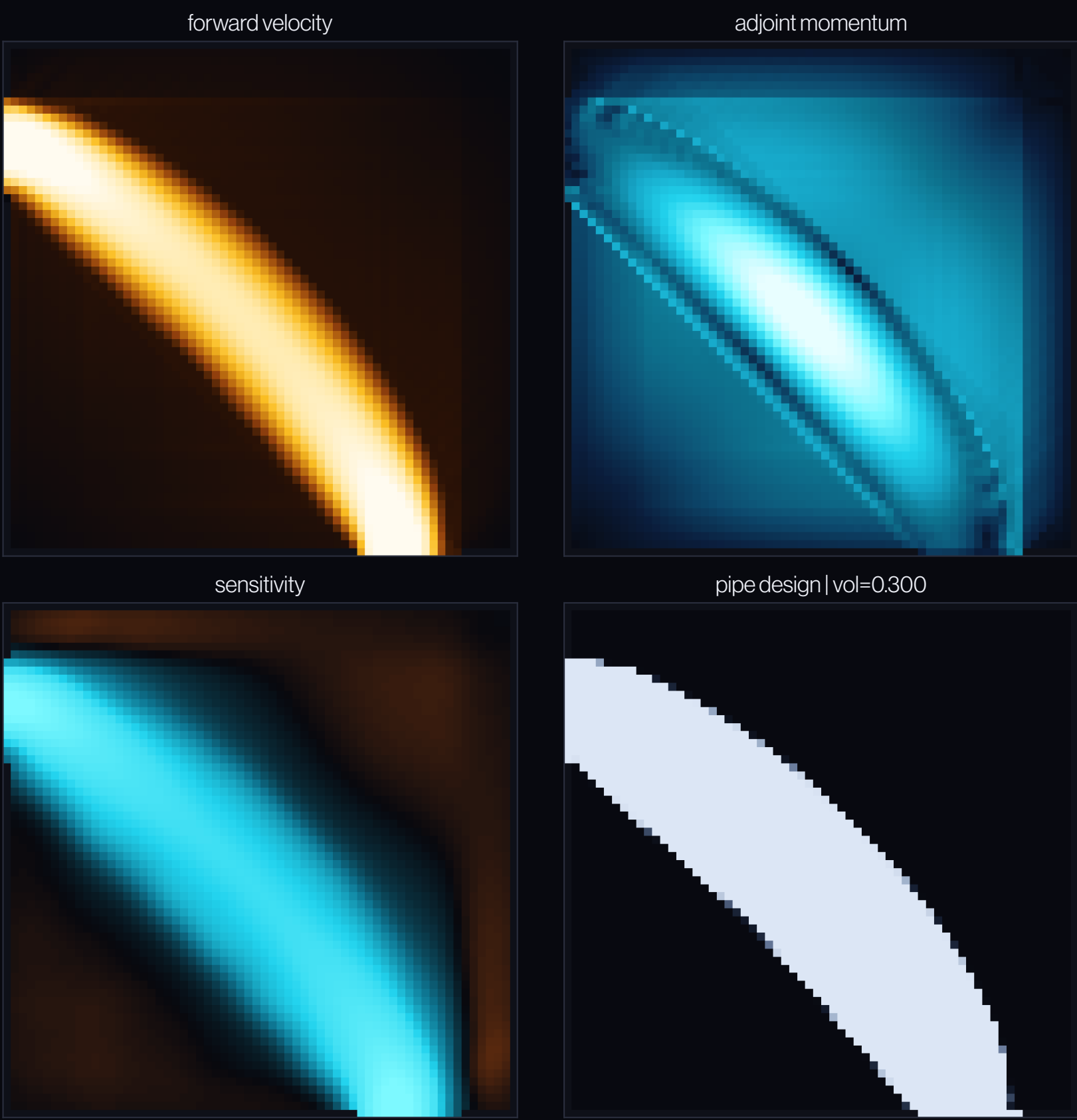
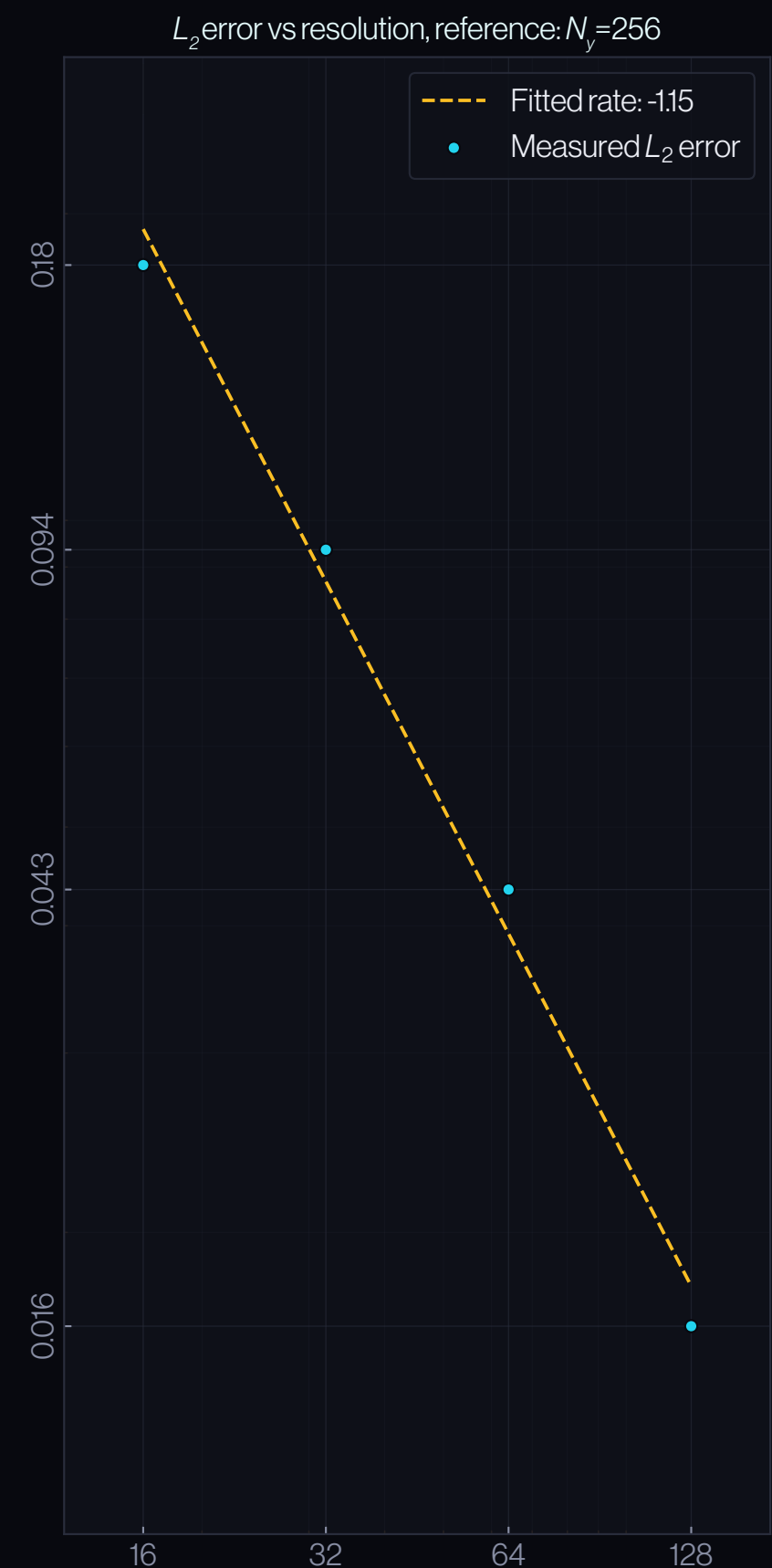
.03

numerical verification

Most fluid-flow scenarios in realistic geometries have no exact solution, so we simulated an obstacle case at five resolutions and measured the error between coarse and fine grids.

The rate falls to -1.0 . Two errors compete: the bulk flow is second-order accurate; the sharp corners, only first-order. At low Re the first-order term dominates.

methods
Constant Re across resolutions, Poiseuille velocity inlet, Zou–He pressure outlet, error measured on the shared fluid cells only



.04

fluid topology optimization

design as a density field
Every cell is a continuous variable from 0.0 to 1.0. The solver interprets it as friction: at 1 the fluid passes freely, at 0 it is brought to rest. Nothing is ever cut away, cells simply become impassable. The pipe is whatever path the fluid is still allowed to take. The continuous field is differentiable.

algorithm
Simulate the flow, solve the adjoint, construct a sensitivity field that describes how changing each cell's density decreases energy dissipation. Iterate until convergence on dissipation and greyness.

adjoint
Constructing a gradient by testing each cell individually is computationally expensive. Using the adjoint method makes the cost of finding sensitivities independent of the total number of design variables. We verified the adjoint gradient matches the finite-difference calculation exactly.

methods
Brinkman penalisation, discrete adjoint, optimality criteria optimizer with volume constraint, sensitivity filter, continuation on Brinkman penalisation and Heaviside projection



.05

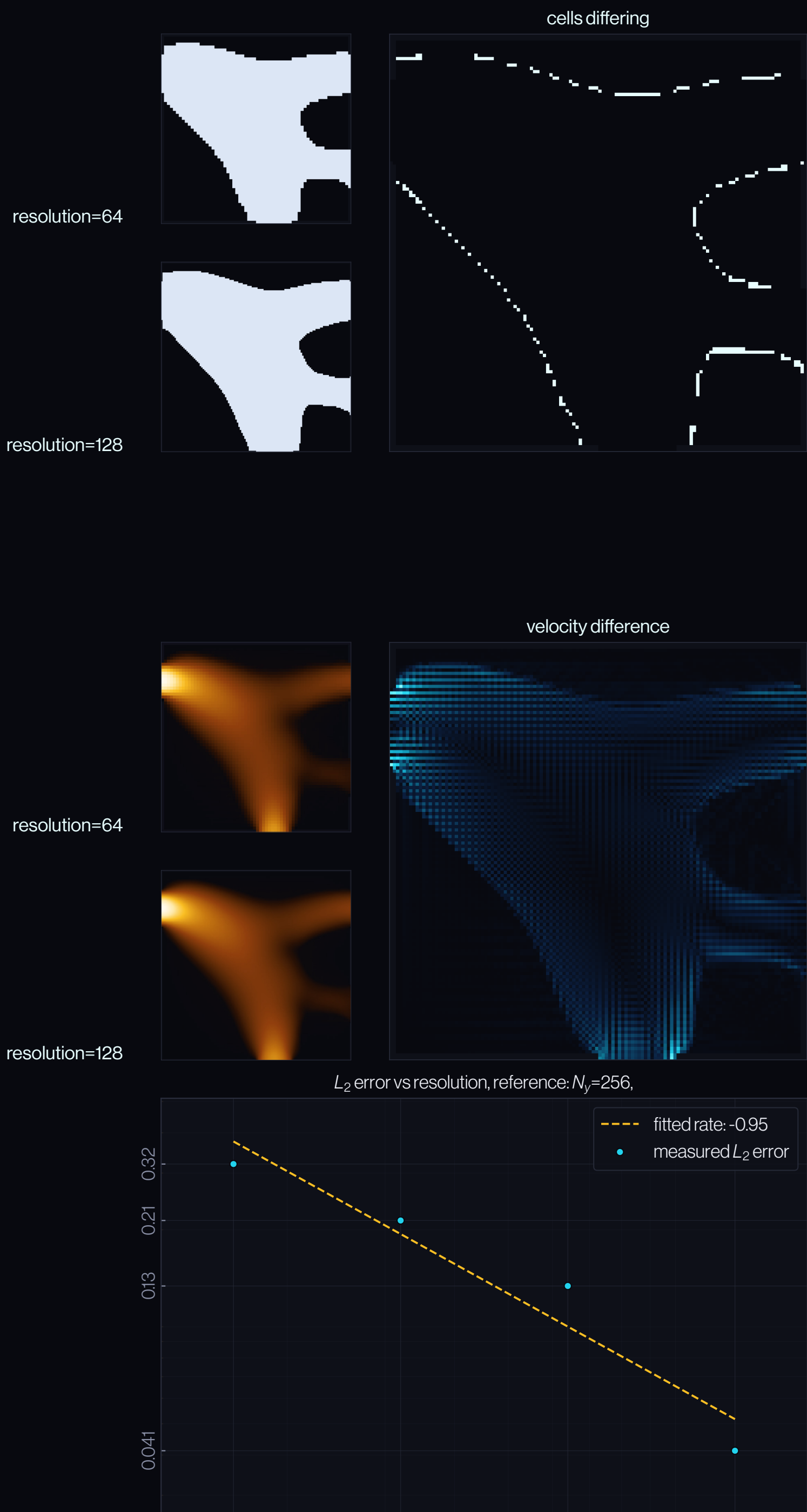
application flow distributor

The problem was inspired by the field of microfluidics, where devices such as lab-on-a-chip systems often need one incoming stream divided between several outlets in fixed proportions.

In our implementation, flow rates are prescribed directly for two outlets, and the third is set by pressure, anchoring the density field to let mass conservation close the balance.

Once the algorithm is ready, the setup is customizable. The plots show the diversity of designs generated with the same solver by setting different outflow proportions and changing outlet walls.

methods
Parabolic velocity inlet, Zou–He velocity outlets set to a fraction of inlet flux, one pressure outlet, adjoint boundary conditions after Luo et al.



.06

mesh independence

Sections .02 and .03 verified the flow solver. To verify the produced design, we ran the same problem at five resolutions and compared each to the finest.

Between 95.3% and 99.3% of cells agree on solid versus fluid, from the coarsest grid to the second-finest.

The disagreement is entirely on the boundaries. Coarse and fine designs place the same channels in the same places. The flow follows – velocity differences sit in the boundary layers and vanish in the channel interior.

The filter radius fixes the smallest feature a design may contain. Held constant as a fraction of the grid, the same physical design appears at every resolution – refined, not reinvented.

conclusion
Given only inlets, outlets, and a material budget, the workflow produces a manufacturable geometry with no shape drawn by hand. The solver has various applications in engineering, including microfluidics. Potential extensions are heat transfer, multiple objectives, and 3D.